Thyll Müller

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Experience

JP Morgan Chase | **Glasgow, Scotland** *Python, GraphQL, JavaScript, React* Jan 2025 - Present

Senior Associate Software Engineer (SWE III)

- Owned the design and delivery of the front-end trade capture UI for JPMorgan's Bond Portfolio TRS Platform migration to Athena, enabling the desk to scale from $10mm to a target $50mm in revenue by replacing error-prone legacy booking workflows (Pyramid/Poseidon) with a modern, resilient interface that reduced trade booking time by 85%.
- Key contributor across the full Bond Portfolio TRS go-live lifecycle, from instrument build through trade capture, pricing, and lifecycling. Delivering a platform that supports significantly higher trading volumes while decommissioning legacy infrastructure and improving downstream reporting accuracy for Operations, Risk and P&L.
- Championed continuous engineering improvement with rigour, driving well-tested refactors such as reducing 1300 line monolithic modules into 10 focused files with comprehensive test coverage, cutting commit contention by 44% and pileup by 58% despite a 67% increase in development activity, unblocking parallel work-streams across the team.
- Managed a summer intern to develop an automation framework for lower-environment trade bookings aiming to accelerate developer and QA testing.

Junior Associate Software Engineer (SWE II) May 2023 - Jan 2025

- Developed and delivered various new bulk trade management workflows for Credit Derivative Flow products within JPMorgan's Athena trade management platform, replacing fragile generic operational processes with dedicated tooling that reduced manual intervention and improved workflow accuracy for trading and operational users.
- Designed and implemented GraphQL APIs, mutations, and frontend view models to support complex trade management workflows, integrating with Athena Trade and Risk systems while enabling responsive, data-intensive UIs built with JavaScript/React.
- Collaborated closely with core Athena platform teams, QR developers, Operations, Front Office, and downstream reporting stakeholders to gather requirements, coordinate cross-system deliveries, and ensure reliable rollout of business-critical trading functionality around Bond Portfolio TRS.
- Provided high-priority production support for Credit Derivative Flow products (CDS and index products) for risk-sensitive trading workflows, rapidly resolving lifecycle and booking issues to maintain accurate intraday risk representation.
- Owned production support for critical quarterly IMM processing events, maintaining stability of lifecycle and cashflow generation workflows for over 400,000 Credit Derivative Flow trades.
- Acted as Technical Lead for a JPMorgan Force for Good engineering team, defining architecture, coordinating delivery across contributors, and overseeing implementation of client-facing software solutions for non-profit organisations.
- Organized and conducted interviews for new hires, ensuring a smooth on-boarding process.

Waracle | **(Remote) Glasgow, Scotland** *Python, Docker, Jinja2, Azure* Sep 2022 - Apr 2023

Junior Python Developer

- Focused on creating and testing data transformation pipeline code at scale.
- Writing unit and integration tests for newly submitted code.
- Built custom APIs to interact with Azure CosmosDB.
- Built and configured docker containers to mimic client infrastructure for strict testing requirements.
- Worked within a large agile team, utilising VCS to implement performant code across the project.

IBM | **(Remote) Leicester, England** *IBM Watson, AWS Connect/Lex/Polly/S3* Mar 2022 - Sep 2022

Conversational Analyst

- Implementing internal and external conversational AI solutions for clients.
- Applied NLP/NLU to improve ML performance in dialog flows for bond trading virtual assistant.
- Liaising with clients to deliver AI solutions to specifications as envisioned by them and guaranteeing successful virtual assistant journeys for customers.

Education

University of Strathclyde Sep. 2021 – Sep. 2022

Data Analytics MSc. (Distinction – 82%)

- Provided data analytics and data science consulting services to clients within organizations such as EY, Capita, and Glasgow City Council.
- Demonstrated proficiency in data science programming using both R and Python.
- Covered a diverse range of modules, including Databases & SQL/NoSQL, Artificial Intelligence, Machine Learning, Deep Learning & Algorithms, Data Analytics with R & Python, as well as Risk Analysis, Financial Econometrics and Business Intelligence
- Specialised in artificial intelligence, machine learning, and deep learning techniques.
- Leveraged cutting-edge frameworks and libraries, including TensorFlow, DBSCAN, Scikit-learn, PyTorch, and various Gradient Boosting Algorithms.

University of Glasgow Sep. 2017 – July 2021

Pharmacology BSc. (Hons) (2:1)

- Gained experience in experimental design and proficient in investigative laboratory techniques.
- Proficient in foundational statistics and analysing biological data using software like R and GraphPad Prism.
- Conducted an Honours Project Dissertation on "Evaluating vitamin D levels as a risk factor for morbidity and mortality rates in SARS-CoV-2."

Technical Skills

Python | *5 Years Of Experience*

- **Frameworks:**

API:	FastAPI, requests, responses
Database & Tree:	AST, GraphQL
Data Manipulation:	Elementtree, Jinja2, JSON, Numpy, Pandas, SQL, SQLAlchemy, XML
Machine Learning:	AdaBoost, PyCaret, Pytorch, Tensorflow, XGBoost
Testing:	PyTest & Unittest

JavaScript | *2 Years Of Experience*

- **Libraries:**

UI Components:	React, React Router, Material-UI, AgGrid
State Management:	Redux, Context API
Other Tools:	Jest, Yarn, Lerna

Virtual Environments | *2 Years Of Experience*

- **Application / Framework:**

Docker:	Proficient in building and managing containerised applications
Python Specific:	pipenv, pyenv, virtualenv